Low-Light Image Enhancement Using Classical Digital Image Processing Techniques: A Comparative Study

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Course: CSE4007 – Digital Image Processing | Academic Year 2025–2026
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***Abstract* - Images captured under low ambient illumination suffer from a compressed dynamic range, weak edge contrast, colour distortion and amplified sensor noise. Although deep-learning enhancers dominate recent benchmarks, classical Digital Image Processing (DIP) techniques remain attractive where deterministic behaviour, zero training data and low computational cost are required. This paper presents a comparative empirical study of four classical enhancement methods: gamma correction accelerated by an 8-bit look-up table, global histogram equalization of the luminance channel in YCrCb space, Contrast Limited Adaptive Histogram Equalization (CLAHE) on the L* channel of CIE-LAB space, and Single-Scale Retinex (SSR). An experimental hybrid pipeline that cascades gamma correction, CLAHE and fast non-local means denoising is also evaluated. All methods are tested on the 15 paired images of the LOL eval15 split using PSNR, SSIM, Shannon entropy, mean brightness, mean contrast and CPU runtime. SSR (σ = 80) obtains the highest mean PSNR (14.9707 dB), whereas gamma correction (γ = 0.4) obtains the highest mean SSIM (0.6196) at 0.0018 s per 400 × 600 image. The hybrid pipeline gives the highest entropy (6.7719) and an SSIM (0.6179) close to that of gamma correction, but it does not exceed the best single methods in PSNR and is the slowest (0.7487 s). Because the PSNR differences among the leading methods are small compared with the scene-to-scene spread, no method is claimed to be uniformly best; the results instead show clear trade-offs between fidelity, structural preservation and speed.**

Keywords *- Digital Image Processing; Low-Light Image Enhancement; Gamma Correction; Histogram Equalization; CLAHE; Single-Scale Retinex; LOL Dataset; PSNR; SSIM.*

I. Introduction

Image acquisition under low ambient light is one of the most common degradation problems in digital imaging. During an exposure of duration Δt, each photosite of the sensor records a signal that can be modelled as

$$S(x,y) = \eta \cdot \Phi(x,y) \cdot \Delta t + \eta_{\text{Poisson}} + \eta_{\text{Gaussian}} \quad (1)$$

where Φ(x,y) is the scene irradiance, η is the quantum efficiency, η_{Poisson} is photon shot noise and η_{Gaussian} represents read-out and thermal noise [1]. In a dark scene Φ is small, so the signal-to-noise ratio (SNR) is low and most of the scene information is quantized into the lowest few intensity levels of an 8-bit image.

Brightening such an image presents a dilemma. Increasing the gain, in the camera or in post-processing, reveals hidden content but amplifies the noise together with the signal, whereas strong spatial smoothing suppresses the noise at the cost of edges and fine texture. An enhancement method therefore has to raise the global illumination, recover local contrast and limit noise amplification at the same time.

Deep-learning enhancers such as RetinexNet [2], Zero-DCE [3] and EnlightenGAN [4] currently dominate public benchmarks. They need training data and considerable compute, and their behaviour is harder to interpret. Classical DIP methods, in contrast, are explicit mathematical operations that need no training data, execute deterministically and are inexpensive to compute, which makes them natural baselines and candidates for resource-constrained pipelines [1]. Knowing how they compare with one another under identical conditions is therefore still useful, both for practical method selection and for understanding the principles on which more complex methods are built.

Problem statement. Given a low-light RGB image and a paired normal-light reference of the same scene, the task is to produce an enhanced image that is close to the reference in fidelity and structure while keeping the computation low. This paper compares four classical methods (gamma correction, global histogram equalization, CLAHE and single-scale Retinex) and one experimental hybrid pipeline on the LOL benchmark [2]. The contributions of this paper are as follows:

- Implementation and evaluation of four classical enhancement methods under one common protocol on the 15 paired LOL eval15 images.
- Formulation of an experimental hybrid pipeline that cascades gamma correction, CLAHE on the L* channel and fast non-local means denoising, and its comparison with the single-method baselines.
- Multi-metric analysis using PSNR, SSIM, Shannon entropy, mean brightness, mean contrast and CPU runtime, including a discussion of why PSNR and SSIM rank the methods differently.
- Visual and histogram analysis of the behaviour of each method, together with an explicit statement of the limitations of the study.

The organization of the paper is as follows. Section II gives the methodology, including the mathematical formulation of each method, the hybrid pipeline and the evaluation metrics. Section III describes the dataset and experimental setup and presents the results. Section IV gives the conclusions, and Section V discusses limitations and future work.

II. Methodology

Five enhancement pipelines are applied to every low-light image of the dataset, and each output is compared with the paired ground truth using the metrics of Section II-F. The workflow of the proposed hybrid pipeline is shown in Fig. 1; the four baselines are applied to the same inputs.

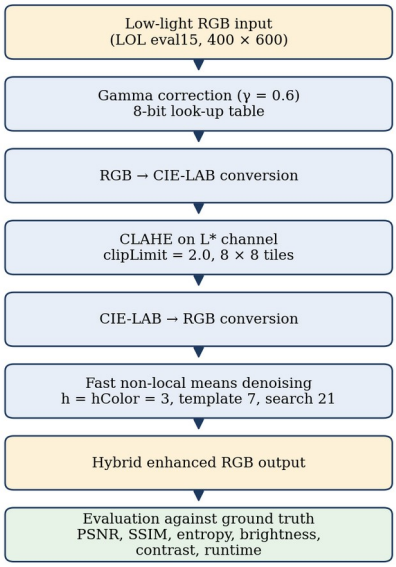


Fig. 1. Workflow of the proposed hybrid pipeline and its evaluation.

A. Power-Law (Gamma) Correction with a Look-Up Table

Gamma correction is a point operation [1] that maps a normalized input intensity r ∈ [0, 1] to an output s ∈ [0, 1] as

$$s = c \cdot r^{\gamma} \tag{2}$$

with c = 1 and γ > 0. Differentiating gives

$$\frac{ds}{dr} = c \cdot \gamma \cdot r^{\gamma - 1} \tag{3}$$

For γ < 1 the slope exceeds one for small r, so narrow dark intensity intervals are stretched over a wider output range while highlights are compressed. For γ > 1 dark values are compressed, and γ = 1 is the identity. The standalone method uses γ = 0.4.

Because 8-bit images take only 256 distinct values, the power function need not be evaluated for every pixel. A look-up table (LUT) is computed once for k = 0, 1, …, 255 as

$$\mathrm{LUT}[k] = \mathrm{clip}\left(255 \cdot \left(\frac{k}{255}\right)^{\gamma}, 0, 255\right) \tag{4}$$

and applied by array indexing (cv2.LUT), so that the per-pixel cost reduces to one memory lookup after a 256-entry precomputation.

B. Global Histogram Equalization in the YCrCb Domain

Histogram equalization (HE) seeks a monotonic mapping T(r) whose output has a uniform probability density over the L gray levels [1]. The continuous mapping uses the cumulative distribution function:

$$s = T(r) = (L-1)\int_0^r p_r(w)\,dw \tag{5}$$

For an M × N image in which n_j pixels have level r_j, the discrete mapping is

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = \frac{L-1}{MN}\sum_{j=0}^{k} n_j \tag{6}$$

Applying HE separately to R, G and B changes the ratios between the channels and produces colour shifts. The image is therefore converted to YCrCb,

$$Y = 0.299R + 0.587G + 0.114B \tag{7}$$

$$\mathrm{Cr} = 0.713(R - Y) + 128 \tag{8}$$

$$\mathrm{Cb} = 0.564(B - Y) + 128 \tag{9}$$

HE is applied to the luminance channel Y only, and the image is converted back to RGB with Cr and Cb unchanged. Because a single mapping is used for the whole image, noisy dark regions are stretched together with informative ones.

C. Contrast Limited Adaptive Histogram Equalization (CLAHE)

Adaptive histogram equalization (AHE) [5] computes mappings from local regions but over-amplifies noise in homogeneous areas. CLAHE [6], [7] limits this effect in three steps. (i) *Tile partitioning:* the image is divided into non-overlapping contextual tiles, here 8 × 8. (ii) *Histogram clipping:* in each tile histogram h(k), the bins above a clip limit β are truncated and the clipped mass is redistributed uniformly over the 256 bins:

$$M_{\mathrm{clip}} = \sum_{k=0}^{255} \max(0, h(k) - \beta) \tag{10}$$

$$h'(k) = \min(h(k), \beta) + \frac{M_{\mathrm{clip}}}{256} \tag{11}$$

In the OpenCV implementation the clipLimit argument (2.0 here) is a normalized value that is converted internally to an absolute limit β per tile. (iii) *Bilinear interpolation:* to avoid visible seams between tiles, each output pixel is obtained by bilinear interpolation of the mappings of the four nearest tile centres, weighted by distance. As with HE, CLAHE is applied only to the L* channel of the CIE-LAB space, leaving a* and b* unchanged.

D. Single-Scale Retinex (SSR)

Retinex theory, introduced by Land and McCann [8] and described in Land's later article [9], models an observed image as the product of illumination L(x, y) and surface reflectance R(x, y):

$$I(x,y) = L(x,y) \cdot R(x,y) \tag{12}$$

Illumination varies slowly across the scene, whereas reflectance carries the intrinsic properties of objects. Taking logarithms turns the product into a difference:

$$\log R(x,y) = \log I(x,y) - \log L(x,y) \tag{13}$$

In the single-scale Retinex of Jobson et al. [10], the illumination is estimated by convolving the image with a normalized Gaussian surround of scale σ:

$$G_\sigma(x,y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{14}$$

The retinex output is then

$$R_{\mathrm{SSR}}(x,y) = \log[I+\varepsilon] - \log[(I+\varepsilon) * G_\sigma] \tag{15}$$

where * denotes convolution and ε = 1.0 prevents log(0). The result is mapped back to [0, 255] by per-channel min–max normalization:

$$R_{\mathrm{out}}(x,y) = 255 \cdot \frac{R_{\mathrm{SSR}}(x,y) - \min R_{\mathrm{SSR}}}{\max R_{\mathrm{SSR}} - \min R_{\mathrm{SSR}}} \tag{16}$$

The scale σ controls the size of the surround: small values emphasize local contrast but can produce halos near strong edges, while large values give smoother illumination estimates and more natural tonal rendering [10]. A large scale, σ = 80, is used in this study.

E. Proposed Hybrid Pipeline

Each classical method has a distinct strength: gamma correction lifts the global illumination, CLAHE strengthens local contrast, and denoising limits the noise that the previous

stages can amplify. The experimental hybrid pipeline (Fig. 1, Table I) cascades these operations. The input is brightened by gamma correction (γ = 0.6), converted to CIE-LAB, and CLAHE (clipLimit = 2.0, 8 × 8 tiles) is applied to L*. After conversion back to RGB, fast non-local means colour denoising [11] is applied with h = 3, hColor = 3, a 7 × 7 template window and a 21 × 21 search window. Non-local means replaces each pixel by a weighted average of pixels whose surrounding patches are similar, so its cost grows roughly with the product of the squared search and template window sizes; this explains the high runtime reported in Section III. The pipeline is a heuristic design with fixed parameters. The role of each stage described above is the design intent; no ablation of the individual stages is reported in this study.

TABLE I. PROPOSED HYBRID ENHANCEMENT PIPELINE

Input: 8-bit RGB image I; γ = 0.6; clipLimit = 2.0; tiles 8 × 8; h = hColor = 3; template 7; search 21
Output: enhanced RGB image I_{out}
1. Build LUT[k] = clip(255(k/255)$^{0.6}$, 0, 255), k = 0…255
2. $I_1 \leftarrow$ LUT(I) // global brightening
3. (L*, a*, b*) $\leftarrow$ RGB→LAB(I_1)
4. L*′ $\leftarrow$ CLAHE(L*; clipLimit, 8 × 8) // local contrast
5. $I_2 \leftarrow$ LAB→RGB(L*′, a*, b*)
6. $I_{out} \leftarrow$ fastNlMeansDenoisingColored(I_2; h, hColor, 7, 21)
7. **return** I_{out}

F. Evaluation Metrics

Every enhanced image is compared with its paired ground-truth image I_{ref}. The mean squared error over all three channels of an M × N image and the peak signal-to-noise ratio are

$$\mathrm{MSE} = \frac{1}{3MN} \sum_{c=1}^{3} \sum_{(i,j)} [I_{\mathrm{ref}} - I_{\mathrm{enh}}]^2 \tag{17}$$

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{255^2}{\mathrm{MSE}} \right) \mathrm{dB} \tag{18}$$

where the sum runs over all pixel positions (i, j) and colour channels c. The structural similarity index (SSIM) [12] compares luminance, contrast and structure of two image regions x and y:

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{19}$$

where μ, σ^2 and σ_{xy} are means, variances and covariance, C_1 = $(0.01 \times 255)^2$ and C_2 = $(0.03 \times 255)^2$. This single-fraction form corresponds to the usual three-term decomposition with $C_3 = C_2/2$. The Shannon entropy of the 256-bin intensity histogram, with probabilities p(k), measures gray-level richness:

$$H = -\sum_{k=0}^{255} p(k) \log_2 p(k) \tag{20}$$

Entropy is a no-reference measure, and a high value can also be caused by noise. Mean brightness is the average pixel intensity of the enhanced image, and mean contrast is taken as the standard deviation of its pixel intensities [VERIFY: confirm these two definitions against your code]. Runtime is measured with time.perf_counter() around the enhancement call only, excluding file I/O, and throughput in frames per second (FPS) is the reciprocal of the mean runtime.

III. EXPERIMENTAL RESULTS

A. Dataset and Experimental Setup

The Low-Light (LOL) dataset [2] contains paired low-light and normal-light photographs of real scenes, captured by varying the exposure time and ISO setting. The official evaluation split, eval15, which contains 15 pairs at 400 × 600 resolution, is used here. An automated verification step confirmed that all filenames match between the low-light and reference folders, that no file is unreadable and that the two images of every pair have identical dimensions. All methods are classical, so no model is trained; the 15 pairs are used only for evaluation. The settings are summarized in Table II.

TABLE II. EXPERIMENTAL SETTINGS

Parameter	Description / value
Dataset	LOL [2], eval15 split
Pairs / size	15 pairs, 400 × 600
Gamma	γ = 0.4, 8-bit LUT (cv2.LUT)
Hist. equalization	Y channel of YCrCb only
CLAHE	L* of CIE-LAB; clipLimit 2.0; 8 × 8 tiles
SSR	σ = 80; ε = 1.0; per-channel min–max
Hybrid	γ = 0.6 → CLAHE (2.0, 8 × 8) → NLM (h = hColor = 3, 7, 21)
Metrics	PSNR, SSIM, entropy, brightness, contrast, runtime
Timer	time.perf_counter(), excludes I/O
Training	None (classical methods)
Parameter selection	[VERIFY: how γ, σ, clipLimit were chosen]
Hardware / software	[VERIFY: CPU, RAM, OS, Python and OpenCV versions]
SSIM settings	[VERIFY: implementation, window, colour handling]
Std. deviation	[VERIFY: population or sample]

B. Quantitative Results

Table III gives the mean and standard deviation of each metric over the 15 scenes, and Figs. 2–4 show the PSNR, SSIM and runtime. SSR (σ = 80) obtains the highest mean PSNR (14.9707 dB), followed by gamma correction (14.5301 dB), HE (14.2356 dB), the hybrid pipeline (13.6609 dB) and CLAHE (9.1323 dB). The SSIM ranking differs: gamma correction (0.6196) and the hybrid pipeline (0.6179) are almost equal, followed by SSR (0.5899), HE (0.4252) and CLAHE (0.3565).

TABLE III. MEAN RESULTS ON LOL EVAL15 (15 IMAGES)

Metric	SSR (σ=80)	Gamma (γ=0.4)	HE (YCrCb)	Hybrid	CLAHE (LAB)
PSNR mean (dB)	**14.9707**	14.5301	14.2356	13.6609	9.1323
PSNR std	3.6755	4.3425	3.1378	4.1031	2.9591
SSIM mean	0.5899	**0.6196**	0.4252	0.6179	0.3565
SSIM std	0.1097	0.1162	0.1018	0.0987	0.1135
Runtime (s)	0.1742	0.0018	**0.0017**	0.7487	0.0065
Throughput (FPS)	5.7	555	**588**	1.3	153
Mean brightness	144.89	73.18	130.79	67.49	29.75
Mean contrast	33.51	25.24	73.64	31.97	16.80
Entropy (bits)	6.7360	6.2015	4.7419	**6.7719**	5.5542

Bold = best value in the row (not applied to brightness and contrast, which have no "better" direction).

The PSNR differences among SSR, gamma correction and HE (0.3–0.7 dB) are small compared with the per-image standard deviations of 3.1–4.3 dB, and the SSIM difference between gamma correction and the hybrid pipeline is only 0.0017. The 15-image experiment therefore does not establish a clear separation among these methods, and no paired significance test was performed. CLAHE is clearly the weakest on both fidelity metrics, which is consistent with its

low mean brightness (29.75): the L* channel is enhanced locally, but the overall exposure remains far from that of the reference.

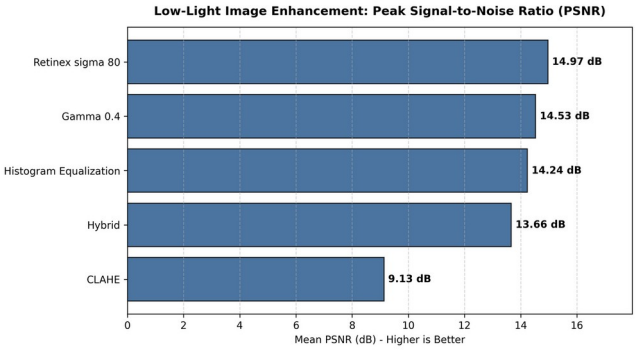


Fig. 2. Mean peak signal-to-noise ratio (PSNR) in dB.

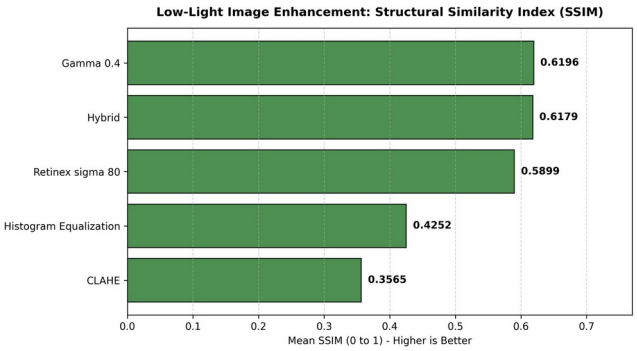


Fig. 3. Mean structural similarity index measure (SSIM).

HE has the highest contrast (73.64) but the lowest entropy (4.7419). A plausible explanation is that equalization of an 8-bit image is a many-to-one mapping that merges neighbouring occupied gray levels and leaves others empty, which can lower entropy even though the histogram spreads over the whole range (see Fig. 6); this was not tested separately. The hybrid pipeline (6.7719) and SSR (6.7360) have the highest entropy, but the difference between them is small.

Runtime per 400 × 600 image is 0.0017 s for HE, 0.0018 s for gamma correction, 0.0065 s for CLAHE, 0.1742 s for SSR and 0.7487 s for the hybrid pipeline (Fig. 4), i.e. approximately 588, 555, 153, 5.7 and 1.3 FPS. These values were measured on a single CPU configuration [VERIFY: hardware].

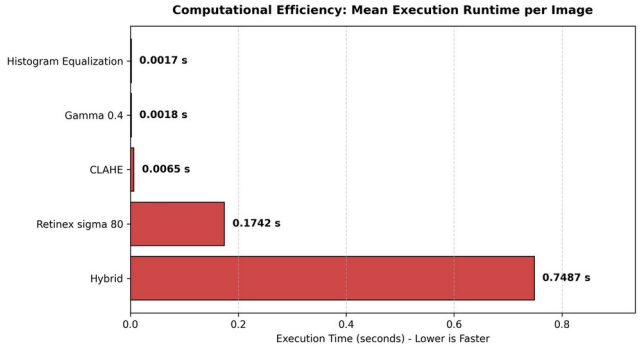


Fig. 4. Mean execution time per 400 × 600 image on a CPU (lower is faster).

C. Per-Scene Results

Table IV is reserved for the PSNR and SSIM values of the individual scenes. [VERIFY: the per-scene table of the project report could not be reconciled with Table III (see accompanying notes), so its values are not reproduced here. Regenerate this table from the per-image output of the evaluation script and fill in the cells.]

TABLE IV. PER-SCENE PSNR (DB) / SSIM

Scene	Gamma	HE	CLAHE	SSR	Hybrid
S1	– / –	– / –	– / –	– / –	– / –
S2	– / –	– / –	– / –	– / –	– / –
S3	– / –	– / –	– / –	– / –	– / –
S4	– / –	– / –	– / –	– / –	– / –
S5	– / –	– / –	– / –	– / –	– / –
S6	– / –	– / –	– / –	– / –	– / –
S7	– / –	– / –	– / –	– / –	– / –
S8	– / –	– / –	– / –	– / –	– / –
S9	– / –	– / –	– / –	– / –	– / –
S10	– / –	– / –	– / –	– / –	– / –
S11	– / –	– / –	– / –	– / –	– / –
S12	– / –	– / –	– / –	– / –	– / –
S13	– / –	– / –	– / –	– / –	– / –
S14	– / –	– / –	– / –	– / –	– / –
S15	– / –	– / –	– / –	– / –	– / –

D. Qualitative and Histogram Analysis

Fig. 5 compares the enhanced outputs for scene 1.png with the input and the reference, and Fig. 6 shows the corresponding RGB histograms. A single scene is shown, so the observations below are illustrative and not statistically conclusive.

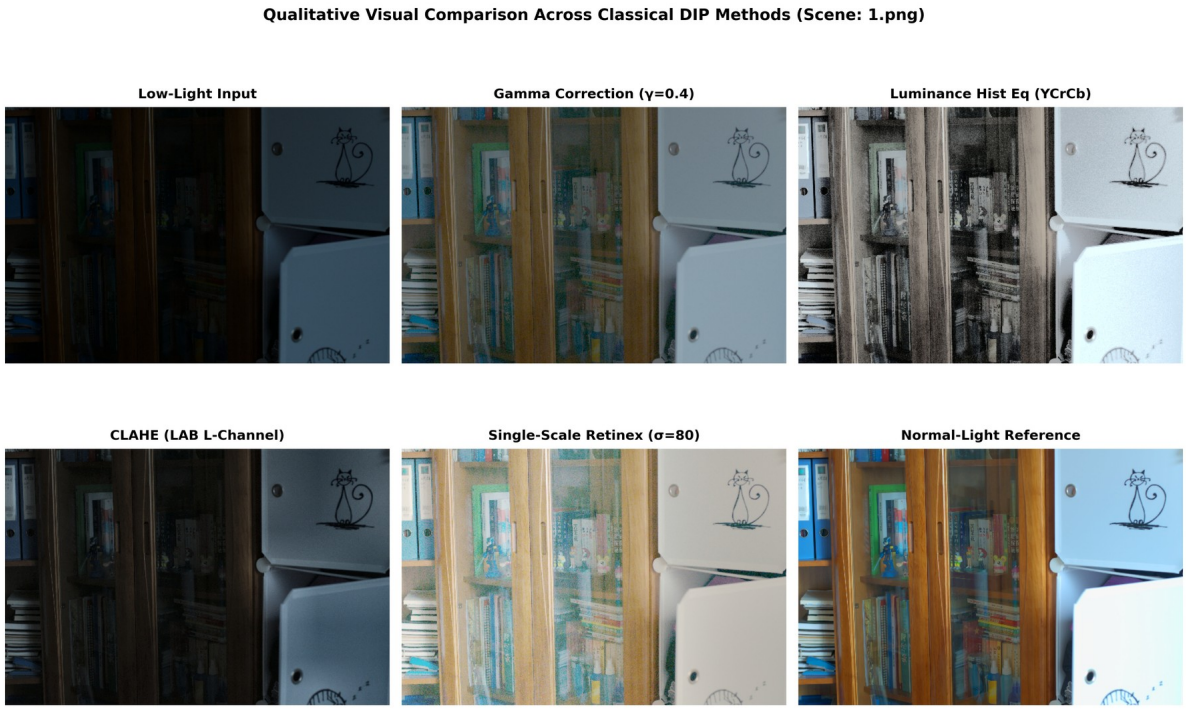


Fig. 5. Qualitative comparison on scene 1.png: low-light input, gamma correction (γ = 0.4), luminance HE (YCrCb), CLAHE (L* of LAB), SSR (σ = 80) and the normal-light reference.

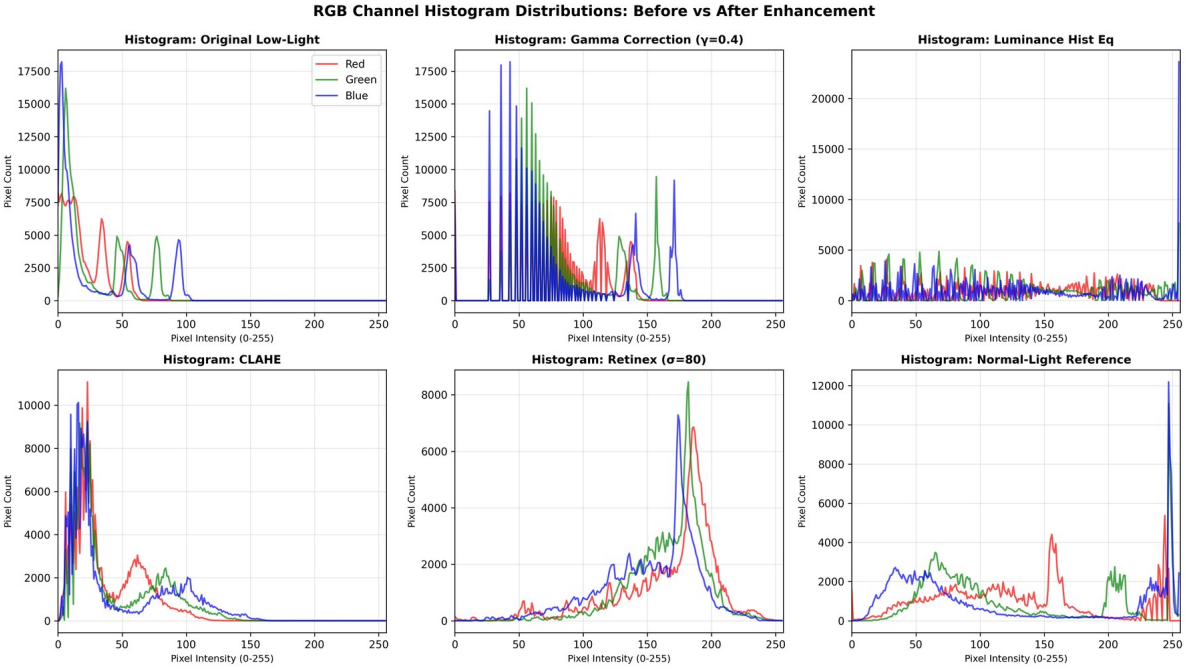


Fig. 6. RGB channel histograms of the input, the four enhanced outputs and the reference, showing dynamic-range stretching and clipping behaviour.

Input. The image is heavily underexposed; shadow textures such as the book spines are barely visible, and the histogram is concentrated at the low end (Fig. 6).

Gamma correction (γ = 0.4). The brightening is uniform and preserves the relative tonal relations and edges without halo artifacts. The output is still darker and less saturated than the reference, and its histogram shows comb-like gaps because the LUT spreads the sparsely occupied input levels over a wider range.

Histogram equalization. Brightness and contrast increase strongly (mean 130.79), but the result looks almost desaturated, dark regions show amplified grain, and the spike at the upper end of the histogram indicates saturated highlights.

CLAHE. Local edges, text and silhouettes are crisper, but the overall brightness remains low (mean 29.75), so the global shadow illumination stays underexposed.

SSR (σ = 80). Shadows and highlights are brightened evenly and the histogram moves to the mid and upper range. In this scene, however, the colours appear less saturated and slightly hazy compared with the reference.

Hybrid pipeline. The hybrid output is not included in Figs. 5 and 6, so no visual evidence for the effect of the non-local means stage is presented here [ADD: generate the hybrid panel and its histogram from your code]. The intended behaviour is that gamma correction lifts the exposure, CLAHE sharpens local detail and denoising suppresses amplified grain; the numerical results of Table III only show that it reaches an SSIM close to that of gamma correction and the highest entropy.

E. Discussion

PSNR versus SSIM. The two metrics rank SSR and gamma correction in opposite order. PSNR depends only on the pixel-wise squared error and penalizes global exposure offsets heavily, whereas SSIM compares local luminance, contrast and structure. SSR separates illumination from reflectance and corrects spatially varying lighting, which can reduce the pixel-wise error over large regions. Gamma correction is a strictly monotonic point mapping that applies no spatial filtering and therefore preserves the ordering of intensities and local structure, which can explain its higher SSIM; the Gaussian surround filtering of SSR can alter local contrast near boundaries. These explanations are consistent with the measured values but were not tested in isolation. HE obtains a PSNR close to that of gamma correction but a much lower SSIM (0.4252 versus 0.6196); its strong stretching of dark noise and its saturated highlights (Fig. 6) are a plausible cause.

Quality–latency trade-off. The methods fall into three groups. HE and gamma correction run in about 2 ms per image (more than 550 FPS on the CPU used) and are the cheapest options. CLAHE (153 FPS) is also fast but, with the settings used, leaves the image too dark. SSR (5.7 FPS) gives the best PSNR at a much higher cost, and the hybrid pipeline (1.3 FPS) is slowest because of non-local means denoising. Whether these figures would carry over to embedded or automotive hardware was not tested in this study.

Overall. No method is uniformly best. SSR is best in PSNR, gamma correction is best in SSIM and is nearly the cheapest, HE is the fastest but gives poor structural similarity and desaturated colour, CLAHE alone does not correct the exposure, and the hybrid pipeline combines near-best SSIM with the highest entropy but is neither best in PSNR nor fast.

IV. Conclusions

This paper compared four classical DIP methods (gamma correction, luminance histogram equalization, CLAHE and single-scale Retinex) and an experimental hybrid pipeline for low-light image enhancement on the 15 paired LOL eval15 images. SSR (σ = 80) achieved the highest mean PSNR (14.9707 dB), and gamma correction (γ = 0.4) achieved the highest mean SSIM (0.6196) with a runtime of 0.0018 s per image. The hybrid pipeline reached an SSIM of 0.6179 and the highest entropy (6.7719) but not the best PSNR, and it was the slowest method (0.7487 s). CLAHE with the chosen settings produced the lowest PSNR and SSIM and left the images dark. The divergence between the PSNR and SSIM rankings shows that a single metric is not sufficient to select an enhancement method, and the runtime results show clear quality–latency trade-offs. Because the differences among the leading methods are small relative to the scene-to-scene variation, the results should be read as a comparison of tendencies rather than as proof of the superiority of any single method.

V. Limitations and Future Work

Several limitations should be acknowledged. The evaluation uses only 15 images from one dataset, and no statistical significance test was performed. Each method was run with fixed parameters (γ = 0.4, σ = 80, clipLimit = 2.0, and the hybrid settings), and no parameter sweep or ablation of the hybrid stages is reported, so the results may not represent the best achievable performance of each technique. Only full-reference metrics, entropy, brightness and contrast were used; no no-reference perceptual metric or noise measure was evaluated, and the hybrid output is not shown in the qualitative figures. Runtime was measured on one CPU configuration in a high-level language and may differ on embedded hardware. Single-scale Retinex can reduce colour saturation, and no deep-learning baseline was included. Future work will therefore include parameter tuning and ablation studies, multi-scale Retinex variants, adaptive choice of γ, paired significance tests, additional metrics and datasets, comparison with learning-based enhancers, and benchmarking on resource-constrained devices.

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